

§ Differential Equations of first order and of higher degree

The equation is of the form

$$f(x, y, p) = 0 \text{ where } p = \frac{dy}{dx}.$$

$\longrightarrow \textcircled{A}$

[The aim is to eliminate p from the equation and establish a relation between x and y]

I. Equations Solvable for p

If eqⁿ $\textcircled{A}$ is of this type, then

- a) Factorize the equation
- b) Make each factor $= 0$ and replace p by $\frac{dy}{dx}$ and solve them to obtain $f_1(x, y) = c$, $f_2(x, y) = c$ etc.

~~c)~~

- c) Solution is given by

$$[f_1(x, y) - c][f_2(x, y) - c] = 0.$$

Ex Solve: $\left(\frac{dy}{dx}\right)^2 + 7\frac{dy}{dx} + 12 = 0 \rightarrow \textcircled{1}$

Solⁿ $\textcircled{1} \Rightarrow p^2 + 7p + 12 = 0$, $p = \frac{dy}{dx}$

$\Rightarrow (p+3)(p+4) = 0$

$\Rightarrow p+3=0$ & $p+4=0$

$\Rightarrow \frac{dy}{dx} + 3 = 0 \quad \left| \Rightarrow \frac{dy}{dx} + 4 = 0 \right.$

$\Rightarrow y+3x=c \quad \left| \Rightarrow y+4x=c \right.$

$\therefore$ Solⁿ is $(y+3x-c)(y+4x-c) = 0$

Ex Solve $\left(\frac{dy}{dx}\right)^2 + \frac{dy}{dx} \left(\frac{y}{x} - \frac{x}{y}\right) = 1 \rightarrow \textcircled{1}$

Solⁿ $\textcircled{1} \Rightarrow p^2 + p\left(\frac{y}{x} - \frac{x}{y}\right) = 1$

$\Rightarrow p^2 + p \frac{y^2 - x^2}{xy} = 1$

$\Rightarrow p^2 xy + p(y^2 - x^2) = xy$

$\Rightarrow p^2 xy + p y^2 - p x^2 - xy = 0$

$\Rightarrow p y (p x + y) - x (p x + y) = 0$

$\Rightarrow (p x + y)(p y - x) = 0$

Now $p x + y = 0 \Rightarrow \frac{dy}{dx} x + y = 0 \Rightarrow \frac{dx}{x} + \frac{dy}{y} = 0$

$\Rightarrow \boxed{xy = c}$

and $p y - x = 0 \Rightarrow \frac{dy}{dx} y - x = 0 \Rightarrow x dx - y dy = 0$

$\therefore$ Solⁿ is $(xy-c)(x^2-y^2-c) = 0$ (2) $\Rightarrow \boxed{x^2 - y^2 = c}$

HW Solve: (1) $p^2 - 2py = 3y^2$

(2) $x^2 p^2 + xy p - 6y^2 = 0$

(3) $p(p+y) = x(x+y)$

(4) $p(p^2 + xy) = p^2(x+y)$.

II Clairaut's Eqⁿ

An eqⁿ of the form

$$\boxed{y = px + f(p)}, \quad p = \frac{dy}{dx}$$

is known as Clairaut's Eqⁿ. If the eqⁿ is of this form, then the solution is given by

$$\boxed{y = cx + f(c)}, \quad \boxed{p \text{ put } p=c}$$

Ex Solve: ① $y = px + \sqrt{1+p^2} \rightarrow$ ①

Solⁿ Since ① is of Clairaut's form the solution is $y = cx + \sqrt{1+c^2}$.

(ii) $(px - y)^2 = p^2$

$\Rightarrow px - y = \pm p \Rightarrow y = px \mp p$.

which is Clairaut's form

$\therefore$ Solⁿ is $\boxed{y = cx \mp c}$

③

HW on Clairaut's Eqⁿ

Solve: (1) $y = px + \frac{a}{p}$

(2) $y = px - 2p^2$

(3) $y = p(x-b) + \frac{a}{p}$

(4) $p = \log(px - y)$

(5) $p = \sin(y - xp)$.

III Eqⁿ solvable for y : The eqⁿ is of the form
$$f(x, y, p) = 0 \rightarrow \textcircled{A}$$

Steps a) Write the eqⁿ $\textcircled{A}$ as

$$y = \phi(x, p) \rightarrow \textcircled{B}$$

(i.e. y in terms of x & p).

(b) Differentiate $\textcircled{B}$ w.r.t x and find

$$\frac{dy}{dx} = \phi_1(x, p, \frac{dp}{dx})$$

$$\Rightarrow p = \phi_1(x, p, \frac{dp}{dx}) \rightarrow \textcircled{C}$$

where $\textcircled{C}$ is an eqⁿ in p and x

(c) Factorize $\textcircled{C}$ and find a solution for p from the factor containing the term $\frac{dp}{dx}$, say $p = \phi_2(x, c) \rightarrow \textcircled{D}$.

(d) Substitute (D) in (A) and obtains the solution. OR eliminate p between (A) and (D) to obtain a relation between x and y .

Ex Solve: $4xp^v - 8yp - x = 0 \rightarrow (1)$

Soln (1) $\Rightarrow 4xp^v - x = 8yp$

$\Rightarrow y = \frac{4xp^v}{8p} - \frac{x}{8p}$

Step (a) $= \frac{1}{2} xp - \frac{1}{8} \frac{x}{p} \rightarrow (2)$

Now $\frac{dy}{dx} = \frac{1}{2} \left[x \frac{dp}{dx} + p \right] - \frac{1}{8} \left[\frac{1}{p} - \frac{x}{p^v} \frac{dp}{dx} \right]$

Step (b) $\Rightarrow p = \frac{1}{2} x \frac{dp}{dx} + \frac{1}{2} p - \frac{1}{8p} + \frac{x}{8p^v} \frac{dp}{dx}$

$\Rightarrow \frac{1}{2} x \frac{dp}{dx} - \frac{1}{2} p + \frac{x}{8p^v} \frac{dp}{dx} - \frac{1}{8p} = 0$

$\Rightarrow \frac{1}{2} \left[x \frac{dp}{dx} - p \right] + \frac{1}{8p^v} \left[x \frac{dp}{dx} - p \right] = 0$

$\Rightarrow \left[x \frac{dp}{dx} - p \right] \left[\frac{1}{2} + \frac{1}{8p^v} \right] = 0$

Step (c) $\Rightarrow x \frac{dp}{dx} - p = 0 \Rightarrow \frac{dp}{p} = \frac{dx}{x}$

$\Rightarrow \log p = \log x + \log c$

$\Rightarrow \boxed{p = cx}$

Putting $p = cx$ in (1), we get the required solution as

$$4x(cx)^2 - 8y(cx) - x = 0$$

$$\Rightarrow 4c^2x^3 - 8cxy - x = 0$$

$$\Rightarrow 4c^2x^2 - 8cy - 1 = 0 //$$

HW Solve: (1) $y + px = p^2x^4$

$$(2) \quad xp^2 + x = 2yp$$

$$(3) \quad y = 2px + p^2$$

$$(4) \quad y = xp^2 + 2xp$$

IV Egⁿ solvable for x : egⁿ is $f(x, y, p) = 0$
 $\downarrow$
 (A) $p = \frac{dy}{dx}$

(a) Write (A) as $x = \phi(y, p)$

(b) Find $\frac{dx}{dy} = \phi_1(y, p, \frac{dp}{dy})$ and put $\frac{dx}{dy} = \frac{1}{p}$.
 to find an egⁿ $\phi_2(y, p, \frac{dp}{dy}) = 0$ $\rightarrow$ (B)

(c) ~~SP~~ Factorize (B) and obtain $p = \phi_3(c, y)$

(d) Put the value of p in (A) and that is the required egⁿ.
 (6)

Hw Solve: (1) $x = \log p \Rightarrow \frac{dx}{dy} = \frac{1}{p} \frac{dp}{dx}$

$\therefore$ Soln is

$$x = \log(x+c) //$$

$$2) \frac{1}{p} = \frac{1}{p} \frac{dp}{dx}$$

$$2) \frac{dp}{dx} = 1$$

$$2) dp = dx$$

$$2) p = x + c.$$

$$(2) \quad 4x = py(p^2 - 3)$$

$$(3) \quad x = py - p^2$$

$$(4) \quad p^2 - xp + y = 0$$

$$(5) \quad xp^2 - yp - y = 0$$

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